

Раздел 1.

su user1

```
root@eltex-practice2-v1:~# su user1
Warning: your password will expire in 6 days.
user1@eltex-practice2-v1:~/root$
```

Подсчитайте количество процессов, имеющих несколько потоков
выполнения

#всё есть awk

```
ps -eL | awk -f p1-1.awk -
```

Proc. using more than 1 thread: 10

#p1-1.awk; сложность в худшем случае $2n$

```
{
    cnt[$1]++
}
END{
    total = 0
    for(p in cnt)
        if(cnt[p]>1)
            total++
    print "Proc. using more than 1 thread:", total
}
```

3. top

f – интерактивный режим

пробел вкл/выкл поле

правая стрелка – захватить поле для перемещения, двигать стрелками

стрелка влево – подтвердить перемещение

```
Fields Management for window 4:User, whose current sort field is USER
  Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
  'd' or <Space> toggles display, 's' sets sort. Use 'q' or <Esc> to end!

* PID      = Process Id          nsPID      = PID namespace Inode
* PPID     = Parent Process pid  nsUSER     = USER namespace Inode
* UID      = Effective User Id   nsUTS      = UTS namespace Inode
* USER     = Effective User Name LXC         = LXC container name
* RUSER    = Real User Name      RSan       = RES Anonymous (KiB)
* TTY      = Controlling Tty     RSfd       = RES File-based (KiB)
* TIME+    = CPU Time, hundredths RSlk       = RES Locked (KiB)
* %CPU     = CPU Usage           RSsh       = RES Shared (KiB)
* %MEM     = Memory Usage (RES)  CGNAME     = Control Group name
* S        = Process Status      NU         = Last Used NUMA node
* COMMAND  = Command Name/Line  LOGID      = Login User Id
* RUID     = Real User Id        EXE        = Executable Path
```

```
top - 11:22:54 up 2 days, 20:22, 3 users, load average: 0.00, 0.00, 0.00
Tasks: 117 total, 1 running, 116 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2708.9 free, 452.7 used, 1020.1 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3463.1 avail Mem
```

PID	PPID	UID	USER	RUSER	TTY	TIME+	%CPU	%MEM	S	COMMAND
21327	21326	1001	user1	user1	pts/0	0:00.02	0.0	0.1	S	bash
21385	21327	1001	user1	user1	pts/0	0:00.02	0.0	0.2	R	top
12536	1	997	systemd+	systemd+	?	0:00.40	0.0	0.2	S	systemd-timesyn
12534	1	992	systemd+	systemd+	?	0:00.22	0.0	0.3	S	systemd-resolve
12533	1	998	systemd+	systemd+	?	0:01.59	0.0	0.2	S	systemd-network
12583	1	103	syslog	syslog	?	0:00.10	0.0	0.1	S	rsyslogd
1	0	0	root	root	?	0:06.12	0.0	0.3	S	systemd
2	0	0	root	root	?	0:00.02	0.0	0.0	S	kthreadd
3	2	0	root	root	?	0:00.00	0.0	0.0	S	pool_workqueue_release
4	2	0	root	root	?	0:00.00	0.0	0.0	I	kworker/R-rcu_g
5	2	0	root	root	?	0:00.00	0.0	0.0	I	kworker/R-rcu_p
6	2	0	root	root	?	0:00.00	0.0	0.0	I	kworker/R-slub_
7	2	0	root	root	?	0:00.00	0.0	0.0	I	kworker/R-netns
12	2	0	root	root	?	0:00.00	0.0	0.0	I	kworker/R-mm_pe
13	2	0	root	root	?	0:00.00	0.0	0.0	I	rcu_tasks_kthread
14	2	0	root	root	?	0:00.00	0.0	0.0	I	rcu_tasks_rude_kthread
15	2	0	root	root	?	0:00.00	0.0	0.0	I	rcu_tasks_trace_kthread
16	2	0	root	root	?	0:00.07	0.0	0.0	S	ksoftirqd/0
17	2	0	root	root	?	0:02.10	0.0	0.0	I	rcu_preempt
18	2	0	root	root	?	0:00.99	0.0	0.0	S	migration/0
19	2	0	root	root	?	0:00.00	0.0	0.0	S	idle_inject/0
20	2	0	root	root	?	0:00.00	0.0	0.0	S	cpuhp/0
21	2	0	root	root	?	0:00.00	0.0	0.0	S	cpuhp/1

Разделение через screen

```
user1@eltex-practice2-v1:~/task3$ passwd
Changing password for user1.
Current password: |
```

и

user1

```
top - 11:39:12 up 2 days, 20:38, 3 users, load average: 0.00, 0.00, 0.00
Tasks: 123 total, 1 running, 122 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2680.7 free, 463.9 used, 1037.3 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3452.0 avail Mem
```

PID	PPID	UID	USER	RUSER	TTY	TIME+	%CPU	%MEM	S	COMMAND
21327	21326	1001	user1	user1	pts/0	0:00.03	0.0	0.1	S	bash
21414	21327	1001	user1	user1	pts/0	0:00.00	0.0	0.1	S	screen
21415	21414	1001	user1	user1	?	0:00.22	0.0	0.1	S	screen
21416	21415	1001	user1	user1	pts/4	0:00.00	0.0	0.1	S	bash
21424	21415	1001	user1	user1	pts/5	0:00.00	0.0	0.1	S	bash
21440	21416	1001	user1	user1	pts/4	0:01.35	0.0	0.2	R	top

у процесса 21424 отличается tty, скорее всего это passwd

к ввод PID, сигнал

MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3450.3 avail Mem

Send pid 21424 signal [15/sigterm] 2

PID	PPID	UID	USER	RUSER	TTY	TIME+	%CPU	%MEM	S	COMMAND
21327	21326	1001	user1	user1	pts/0	0:00.03	0.0	0.1	S	bash
21414	21327	1001	user1	user1	pts/0	0:00.00	0.0	0.1	S	screen
21415	21414	1001	user1	user1	?	0:00.27	0.0	0.1	S	screen
21416	21415	1001	user1	user1	pts/4	0:00.00	0.0	0.1	S	bash
21424	21415	1001	user1	user1	pts/5	0:00.00	0.0	0.1	S	bash
21440	21416	1001	user1	user1	pts/4	0:03.95	1.0	0.2	R	top

после отправки всех сигналов, видимо, мимо

```
user1@eltex-practice2-v1:~/task3$ passwd
Changing password for user1.
Current password:
```

возможно было убито окно утилиты screen

```
No other window.
```

после запуска в другом терминале по тому же принципу (отличный tty),
был убит уже нужный(?) процесс

```
10.07.2026-05:31
Last login: Sun Jul 12 11:37:39 2026 from 172.16.8.4
root@eltex-practice2-v1:~# su user1
Warning: your password will expire in 6 days.
user1@eltex-practice2-v1:/root$ passwd
Changing password for user1.
Current password: Killed
root@eltex-practice2-v1:~#
^[[A^[[Bpasswd: Authentication token manipulation error
passwd: password unchanged
```

```

passwd: password unchanged
^C^C

root@eltex-practice2-v1:~# ^C

root@eltex-practice2-v1:~#
s: command not found
root@eltex-practice2-v1:~#
Command 'dvm' not found, did you mean:
  command 'dvc' from snap dvc (3.59.1)
  command 'dnf' from deb dnf (4.14.0-4.1)
  command 'df' from deb coreutils (9.4-3ubuntu6.2)
  command 'duf' from deb duf (0.8.1-1ubuntu0.24.04.3)
See 'snap info <snapname>' for additional versions.
root@eltex-practice2-v1:~#

```

Внимательнее смотрим в документацию и замечаем L — поиск по команде

```

top - 12:17:47 up 2 days, 21:17, 5 users, load average: 0.00, 0.00, 0.00
Tasks: 127 total, 1 running, 126 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2675.5 free, 468.6 used, 1037.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used, 3447.3 avail Mem

  PID  PPID  UID  USER   RUSER  TTY      TIME+  %CPU  %MEM  S  COMMAND
 22014 22005    0 root    user1   pts/5    0:00.00  0.0   0.1  S  passwd
 12532    1   991 polkitd polkitd ?         0:01.96  0.0   0.2  S  polkitd
   691    1   101 message+ message+ ?         0:00.83  0.0   0.1  S  dbus-daemon

```

```

^ Making proxy SSH connection to 217.71.138.1 port 44556
Making primary SSH connection to 172.16.9.175
Using username "root".
root@172.16.9.175's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Jul 12 12:14:32 PM UTC 2026

System load:  0.0          Processes:      125
Usage of /:   43.7% of 14.66GB  Users logged in:  0
Memory usage: 8%          IPv4 address for ens18: 172.16.9.175
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

76 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Никита Абрамский
10.07.2026-05:31
Last login: Sun Jul 12 12:14:32 2026 from 172.16.8.4
root@eltex-practice2-v1:~# su user1
Warning: your password will expire in 6 days.
user1@eltex-practice2-v1:/root$ passwd
Changing password for user1.
Current password:

```

на сигнале 9 процесс умер окончательно

```
top - 12:20:30 up 2 days, 21:19, 5 users, load average: 0.00, 0.00, 0.00
Tasks: 125 total, 1 running, 124 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 96.3 id, 3.5 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 2675.5 free, 468.5 used, 1037.9 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used, 3447.3 avail Mem

  PID  PPID  UID USER  RUSER  TTY      TIME+  %CPU  %MEM  S COMMAND
  22022      2    0 root    root    ?        0:00.00  0.0   0.0  I kworker/1:0
  12532      1   91 polkitd polkitd ?        0:01.96  0.0   0.2  S polkitd
    691      1  101 message+ message+ ?        0:00.83  0.0   0.1  S dbus-daemon

Making proxy SSH connection to 217.71.138.1 port 44556
Making primary SSH connection to 172.16.9.175
Using username "root".
root@172.16.9.175's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Jul 12 12:14:32 PM UTC 2026

System load: 0.0          Processes: 125
Usage of /: 43.7% of 14.66GB Users logged in: 0
Memory usage: 8%          IPv4 address for ens18: 172.16.9.175
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

76 updates can be applied immediately.
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To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Никита Абламский
10.07.2026-05:31
Last login: Sun Jul 12 12:14:32 2026 from 172.16.8.4
root@eltex-practice2-v1:~# su user1
user1@eltex-practice2-v1:/root$ passwd
Changing password for user1.
Current password: Killed
user1@eltex-practice2-v1:/root$
user1@eltex-practice2-v1:/root$
user1@eltex-practice2-v1:/root$
```

и терминал работает как положено

```
user1@eltex-practice2-v1:/root$ vim ~/task3/file.txt
```

```
[1]+  Stopped                  vim ~/task3/file.txt
```

```
user1@eltex-practice2-v1:/root$ sleep 600
```

```
^Z
```

```
[2]+  Stopped                  sleep 600
```

```
user1@eltex-practice2-v1:/root$ jobs
```

```
[1]-  Stopped                  vim ~/task3/file.txt
```

```
[2]+  Stopped                  sleep 600
```

```
user1@eltex-practice2-v1:/root$ sleep 600 &
```

```
[4] 22070
```

```
user1@eltex-practice2-v1:/root$ renice 10 22070
```

```
22070 (process ID) old priority 0, new priority 10
```

PID	PPID	UID	USER	RUSER	TTY	TIME+	%CPU	%MEM	S	COMMAND	NI
21415	1	1001	user1	user1	?	0:00.39	0.0	0.1	S	screen	0
21416	21415	1001	user1	user1	pts/4	0:00.00	0.0	0.1	S	bash	0
21440	21416	1001	user1	user1	pts/4	0:10.24	0.3	0.2	S	top	0
21996	21995	1001	user1	user1	pts/0	0:00.00	0.0	0.1	S	bash	0
22005	22004	1001	user1	user1	pts/5	0:00.01	0.0	0.1	S	bash	0
22019	21996	1001	user1	user1	pts/0	0:00.81	0.0	0.2	R	top	0
22060	22005	1001	user1	user1	pts/5	0:00.01	0.0	0.3	T	vim	0
22063	22005	1001	user1	user1	pts/5	0:00.00	0.0	0.1	T	sleep	0
22064	22005	1001	user1	user1	pts/5	0:00.00	0.0	0.1	T	sleep	0
22070	22005	1001	user1	user1	pts/5	0:00.00	0.0	0.1	S	sleep	10

top - 12:38:06 up 2 days, 21:37, 5 users, load average: 0.00, 0.00, 0.00											
Tasks: 127 total, 1 running, 123 sleeping, 3 stopped, 0 zombie											
%Cpu(s): 0.1 us, 0.2 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st											
MiB Mem : 3915.9 total, 2669.6 free, 466.8 used, 1045.5 buff/cache											
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3449.1 avail Mem											
PID	PPID	UID	USER	RUSER	TTY	TIME+	%CPU	%MEM	S	COMMAND	NI
22070	22005	1001	user1	user1	pts/5	0:00.00	0.0	0.1	S	sleep	10
22083	21996	1001	user1	user1	pts/0	0:00.03	0.0	0.2	R	top	0
12536	1	997	systemd+	systemd+	?	0:00.41	0.0	0.2	S	systemd-+	0

top - 12:42:07 up 2 days, 21:41, 5 users, load average: 0.00, 0.00, 0.00											
Tasks: 129 total, 1 running, 125 sleeping, 3 stopped, 0 zombie											
%Cpu(s): 0.2 us, 0.0 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st											
MiB Mem : 3915.9 total, 2672.8 free, 463.6 used, 1045.5 buff/cache											
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3452.3 avail Mem											
PID	PPID	UID	USER	RUSER	TTY	TIME+	%CPU	%MEM	S	COMMAND	NI
22060	22005	1001	user1	user1	pts/5	0:00.01	0.0	0.3	T	vim	0
22063	22005	1001	user1	user1	pts/5	0:00.00	0.0	0.1	T	sleep	0
22064	22005	1001	user1	user1	pts/5	0:00.00	0.0	0.1	T	sleep	0
22070	22005	1001	user1	user1	pts/5	0:00.00	0.0	0.1	S	sleep	0
22083	21996	1001	user1	user1	pts/0	0:00.24	0.0	0.2	R	top	0

fg 1

```
user1@eltex-practice2-v1:/root$ jobs
[1]  Stopped                  vim ~/task3/file.txt
[2]-  Stopped                  sleep 600
[3]+  Stopped                  sleep 600
[4]   Running                  sleep 600 &
user1@eltex-practice2-v1:/root$ fg 1
vim ~/task3/file.txt
[4]   Done                     sleep 600
user1@eltex-practice2-v1:/root$ |
```

ВЫХОД ИЗ vim на :q

```
user1@eltex-practice2-v1:/root$ kill -15 22063
user1@eltex-practice2-v1:/root$ jobs
[2]-  Stopped                  sleep 600
[3]+  Stopped                  sleep 600
```

```
trap 'echo Меня голыми руками не возьмёшь!' SIGINT SIGQUIT
```